Cover Letter — Journal of Chemical Information and Modeling

Dear Editors,

We submit the manuscript entitled *"From African-Natural-Product Chemical Space to Resistance-Resilience Hypotheses: An Integrated Computational Study of Antimalarial Polypharmacology"* for consideration as a research article.

The manuscript presents an integrated computational study connecting African-natural-product-inspired chemical-space exploration with target breadth and mutation-aware resistance hypotheses. Its central methodological premise is that these properties must not be collapsed into a single score: target breadth is represented by a target-specific docking profile, whereas resistance resilience is computed separately from mutation panels using a per-target denominator that excludes non-binding wild-type baselines.

The workflow combines a hybrid library of 65 856 molecules with a computationally prioritized set of 19 913 synthesizability-filtered leads and a locked 17-member Set-C cohort. The cohort was evaluated against PfDHFR, PfCRT, PfClpP, and PfATP4 using target-specific structural anchors, yielding 68 raw Vina records that passed an automated geometric gate; the underlying poses remain subject to independent structural review. Because Vina scores are not calibrated across unlike receptor pockets, the manuscript reports target-wise evidence and does not use a cross-target arithmetic mean. Corrected RRS analysis was derived separately from the WT/mutant panel in `docking_mutants.csv`, not from the four-target V5 WT matrix, and classified the cohort as six A*, five B, five C, and one D profiles. The cross-metric analyses were retained as exploratory and null findings were reported without reinterpretation.

The paper's contribution is therefore not a claim of confirmed multi-target inhibition. It is a reproducible computational framework that separates chemical-space novelty, target-profile hypotheses, and mutation resilience; corrects the previous PP-11 target-mixing artifact; and makes structural-evidence classes explicit. The automated integrity audit found no file, schema, cohort, or numerical-recomputation errors, while the independent-review status remains transparently pending. The study does not claim measured potency, confirmed target engagement, resistance circumvention, or experimental IC_{50}/EC_{50}. These calculations define testable priorities for biochemical, cellular, mutant-panel, and molecular-dynamics follow-up.

Source code, candidate manifests, receptor and ligand files, raw Vina outputs, RRS tables, configuration files, and provenance records are available at `https://github.com/NanaEngo/Malaria_codesV2`. All authors have reviewed and approved the manuscript and declare no competing financial interest. This manuscript is not under consideration elsewhere.

Thank you for considering our work.

Sincerely,
Serge Guy Nana Engo
for all authors